

Modelling the behaviour of shape memory materials using machine learning approaches

AE 419 Supervised Learning Project

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Acceptance Certificate

The project report entitled “Modelling the behaviour of shape memory materials using machine learning approaches” submitted by Chauhan Yash Pragneshbhai (180010019) is accepted for evaluation.

Prof. Susmita Naskar

19th April, 2021

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Abstract

Shape memory alloys are capable of delivering advantageous solutions to a wide range of engineering-based problems. Implementation of these solutions, however, is often complicated by the hysteretic, non-linear, thermo-mechanical behavior of the material. Existing models are largely capable of accurately describing this unique behavior, but they require prior characterization of material parameters. Current characterization procedures necessitate extensive data collection and data processing, creating a high barrier of entry for shape memory alloy application. In This project I intend to develop an understanding about the approaches for characterization in which a form of computational intelligence is applied to the task of shape memory alloy parameter characterization. Specifically, I conducted review of a methodology in which the supervised learning model is trained to identify transformation temperatures and stress influence coefficients of shape memory alloy specimens using strain temperature coordinates as inputs. Additionally review of numerical implementation method for Training data generation is also performed through the use of an existing shape memory alloy constitutive model. This project assesses a novel Artificial neural network based method of shape memory alloy (SMA) behaviour characterization. The intent here is to establish an understanding about ML based SMA characterization approaches so as to develop ML model to facilitate and supplement the existing SMA material parameter characterization process, allowing more rapid characterization while requiring less data and SMA expertise.

Chapter 1

Introduction

The Shape Memory Materials are stimuli-responsive smart materials with the ability to undergo a large recoverable deformation upon the application of an external stimulus like heat, electricity, light, magnetism, moisture, change in pH value

Shape Memory Materials are used in numerous commercial fields such as in consumer products and industrial applications, structures and composites, automotive, aerospace, mini actuators and micro-electromechanical systems (MEMS) and robotics.

Shape memory alloy (SMA) was first discovered by Arne Olander in 1932 and the term "shape-memory" was first described by Vernon in 1941 for his polymeric dental material [9]. The importance of shape memory materials (SMMs) was not recognised until William Buehler and Frederick Wang revealed the shape memory effect (SME) in a nickel-titanium (NiTi) alloy in 1962. [10] Since then, the demand for SMAs for engineering and technical applications has been increasing in numerous commercial fields

Though largely capable of accurately modeling SMA behavior, the constitutive models documented in the literature inherently depend on the accurate characterization of the material typically in the form of identifying values for material parameters.

Developing a means of consistently and accurately identifying these unique SMA physical parameters is not a novel task; different methodologies have been studied. accurate identification of characteristic SMA material parameters necessitates extensive data collection.

Although these methods allow one to identify SMA material properties with some degree of accuracy, these existing methods are currently time-consuming procedures requiring both extensive data collection and processing. Furthermore, these processes typically require some degree of SMA expertise. This creates a somewhat high barrier of entry for both engineers hoping to use the material and researchers desiring to further model advanced behavior of the material.

In this project I conducted study of various modeling approaches and a detailed review of Numerical implementation of SMA thermo-mechanical model using return mapping algorithms with an objective of carrying the work towards building machine learning model Shape memory material based actuators' behaviour characterization.

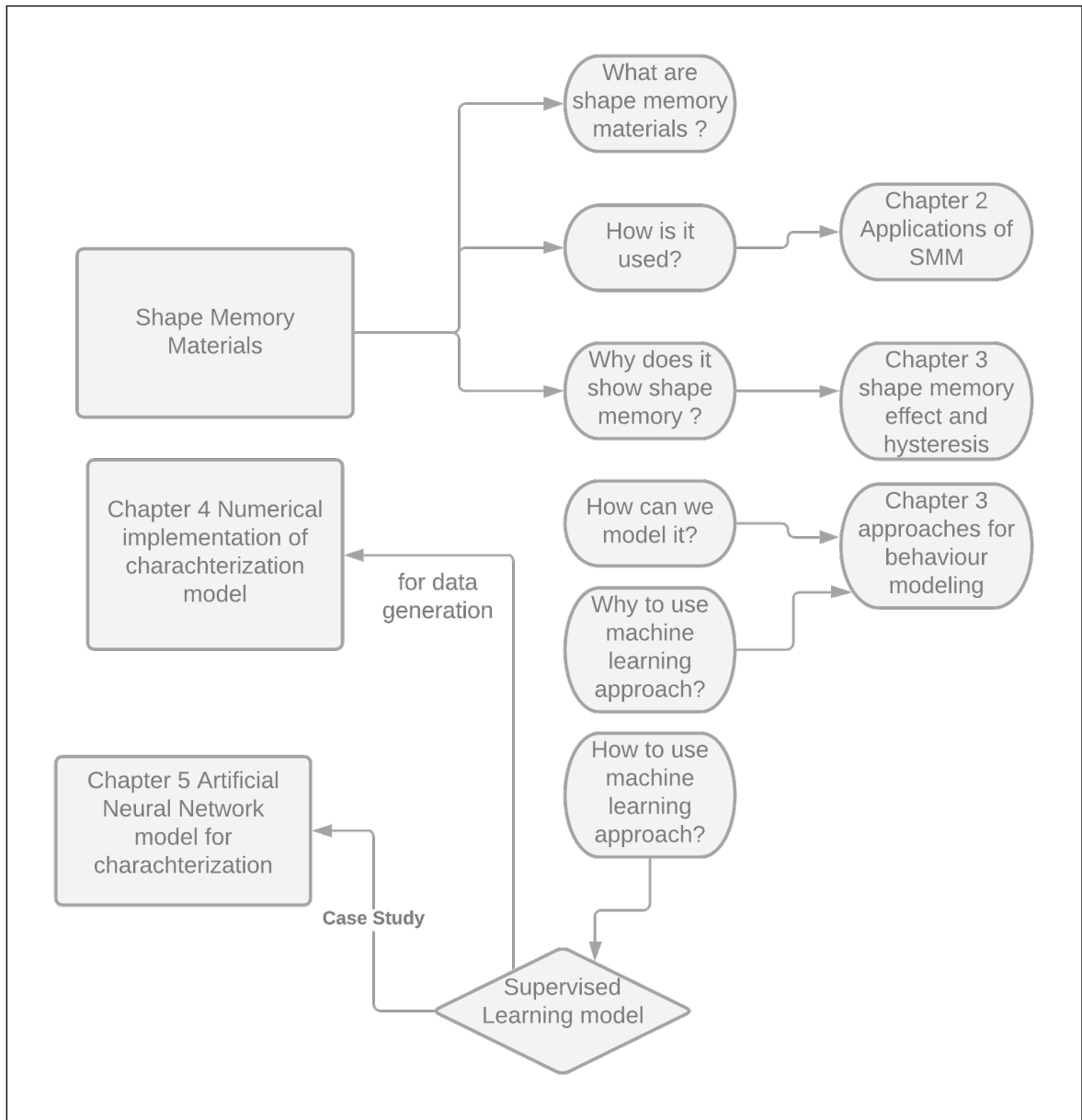


Figure 1.1: Report Layout

Chapter 2

Shape memory materials and its applications

Before proceeding with the study of the behaviour and characterization of Shape Memory Materials its worthwhile to understand the applications of it.

In this chapter I reviewed some of the examples of industrial applications of SMM (Shape Memory Materials). Focusing on the application where Shape memory effect is used for the actuation purposes, and the last section on SMA (Shape Memory Alloy) actuator where I have compared functioning of SMA actuator with other types of actuation methods.

Generally the shape memory applications can be divided into four categories which are according to its primary function of the memory element; The four categories being,

1. Free Recovery - Sole purpose is to cause motion or strain on the applications
2. Constrained Recovery - The memory element is constrained from changing shape and thereby generates a stress or force on the applications
3. Actuator - There is motion against a stress and thus work is bring done by the memory element on the applications
4. Super Elasticity - The applications are isothermal in nature and involve the storage of potential energy

2.1 Aerospace Applications

The use of smart materials such as SMAs gives the chance to design mechanical and aerospace structures for better system performance, such as low vibration, shape control, and structural health monitoring. SMAs are used in morphing applications in different parts of the aircraft such as Engines, wing morphing in which different morphed configurations are possible with the use of Shape memory actuators.

One of the most well-known examples are fixed-wing programs are the Smart Wing program and the Smart Aircraft. This Program implemented active materials, such as SMAs, to optimise the performance of lifting bodies.

Boeing has developed an active serrated aerodynamic device with SMA actuators, which is also known as a variable geometry chevron (VGC) and has been installed on a GE90-115B jet engine (for the Boeing 777-300 ER commercial aircraft) shown in figure 2.1. This device has proven to be very effective in reducing noise during take-off by maximising the chevron deflection.

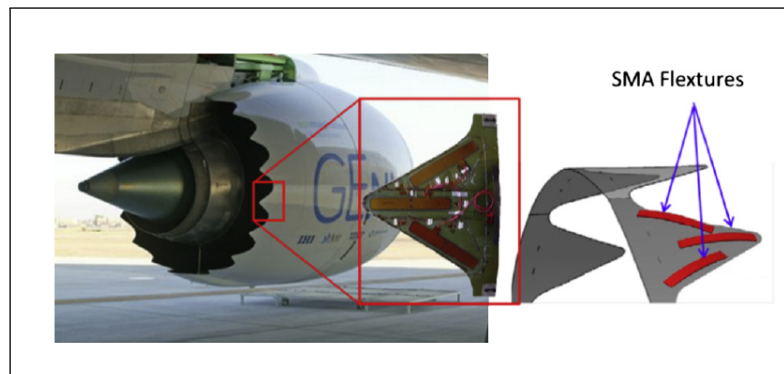


Figure 2.1: Boeing's variable geometry chevron [1]

Sofla et al. [2] have provided a comprehensive review of aircraft wing morphing technologies, and they have also developed a shape morphing wing design for small aircraft by applying antagonistic SMA-actuated flexural structural forms as observed in Figure 2.2, that enable the changing of the wing profile by bending and twisting, to improve the aerodynamic performance

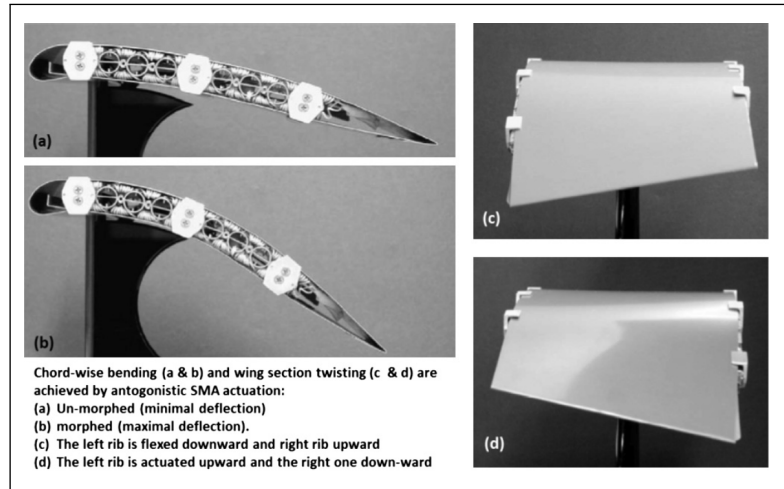


Figure 2.2: Wing Morphing with SMA Actuators [2]

2.2 Bio-Medical Applications

[11] SMA technology has been explored extensively during the past few decades. The properties of Shape memory actuators are suitable for successful implementation in variety of dental applications and since the discovery it has been used for dental applications. key dental application for SMAs involves the use of Nitinol drills used in root canal surgery, which involves careful drilling within the tooth. The advantage of these Nitinol drills is that they can bend to rather large angles, which induce large strains, yet still withstand the high cyclic rotations. Other Medical Applications include Cardiovascular and Orthopaedic, surgery - colorectal, otolaryngology, neurosurgery, ophthalmology, urology and physiotherapy.

SMA device used, exploited successfully the intricate spaces of the surgical environment inside the body with the help of shape memory actuators' unique properties several methods for operation and cure is now possible. Among SMAs, In particular, Nitinol show very unusual material features including bio compatibility that have opened up the doors to many innovative applications in the biomedical field [3]

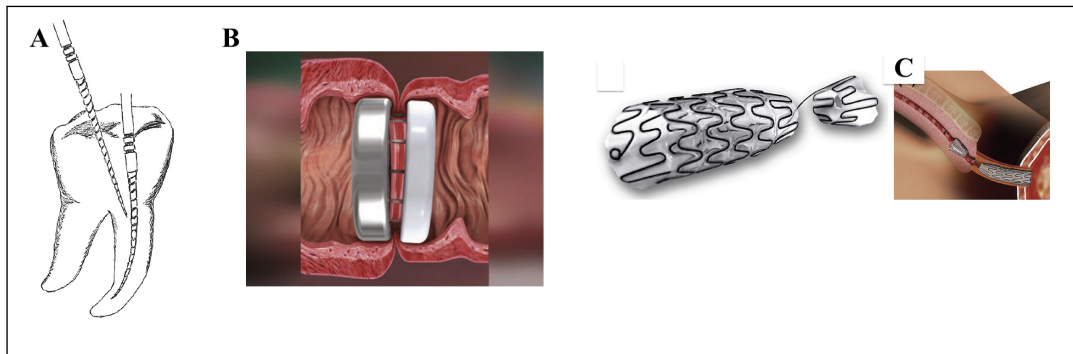


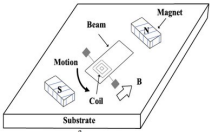
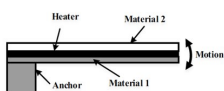
Figure 2.3: Various Bio-Medical Applications of SMA [3] [4] [5]

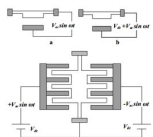
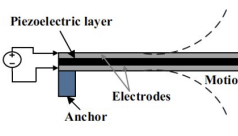
The figure 2.3. figure A shows nitinol dental drills [3], figure B shows colon ring anastomosis device by InnoMedicus [4], figure C shows urethral stent by Allium Medical Ltd [5]

2.3 SMA Actuators

Due to similarity in performing the operation of sensing, controlling and actuating, for the design of SMA based actuators MEMS based actuators and their types are studied. The design and the sizing of an actuation system are performed on the basis of specifications that the actuator itself, as part of an integrated system, has to meet. Among these requirements the intensity of the forces and of the displacements to be transmitted, the frequency bandwidth, the necessary power supply, the weight, the size, and the cost are crucial. [12]

Chapter 2. Shape memory materials and its applications

	Working Principle	Advantages and Disadvantages	Graphic
Electromagnetic	Lorentz Force; $\vec{F}_L = L\vec{I} \times \vec{B}$	Large displacements, rapid response, high accuracy, easy controllability, high reproducibility Difficult to fabricate, high power consumption	
Electrothermal	Difference in thermal expansion of different layers of materials $F_{th} = \alpha_T A E (T - T_0)$	Large displacements, ease of fabrication, large force Requires Large amount of current, low voltage amplifier, high power consumption due to joule heating	

	Working Principle	Advantages and Disadvantages	Graphic
Electrostatic	Electrostatic actuation is based on Coulomb's law which depicts the reciprocal force between two charges with a certain distance that generated between fixed and movable plates.	Has high-frequency response and low power Consumption Need elevated driving voltage & nonlinearity	
Piezoelectric	A piezoelectric actuator defined as transducers that converts an applied potential into a mechanical movement or strain based on the piezoelectric influence	high output force, good operating bandwidth, very compact size, lightweight, low power consumption, and fast response Small displacements, high temperature sensitivity, <u>non linear</u> working, hysteresis	

Here in the table, L = length vector, B = magnetic field, α_T =coefficient of thermal expansion, E = Electric Field and A = Cross section area of beam,
 T = Temperature

MEMS-based devices could be actuated by applying a sinusoidal force $F(t)$, through one of the above mentioned actuating methods

$$M\ddot{\gamma} + b\dot{\gamma} + k\gamma = F(t) \quad (2.1)$$

Equation mentioned above is used to find the general equation of motion where M is the total mass of the top plate, b is the damping coefficient, k spring constant of the flexible beam

The figure below shows an example of SMA actuator i.e, Nitinol on platinum MEMS SMA actuator which is a bi-morph actuator. With functioning similar to piezoelectric actuator. The key feature of a shape memory actuator is it uses only the phase transformation of the alloy for producing force. The strain path, or type of stroke depends on the processed shape of the actuating element.

This allows a large flexibility of design. SMA actuators can work in tension, compression, bending, rotation or a combination of some of those deformation modes, such as mainly applied in springs. [6]

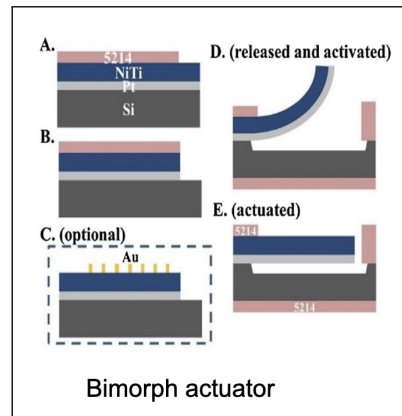


Figure 2.4: Micro Shape Memory Alloy Actuator [6]

Thus with the establishment of an understanding on how shape memory materials are used, especially as an actuator next step is to understand the material science and physical aspects behind the shape memory effect.

Chapter 3

Shape memory effect and approaches for its behaviour modeling

In this chapter, material science and physical aspects related to shape memory are discussed. In the next section summary of literature review of different Shape memory material behaviour modeling approaches and its problems and flaws are discussed. In the end machine learning based modelling approach is described and its implementation is explored in next chapters.

3.1 Shape Memory Effect and Hysteresis

Shape Memory refers to materials with one way, two-way or multi-shape memory effect. Two-way shape memory effect is a reversible transition between two different shapes of the material one of these occurring when the temperature is below transition temperature while the other shape is obtained by increasing the temperature above the Transition temperature.

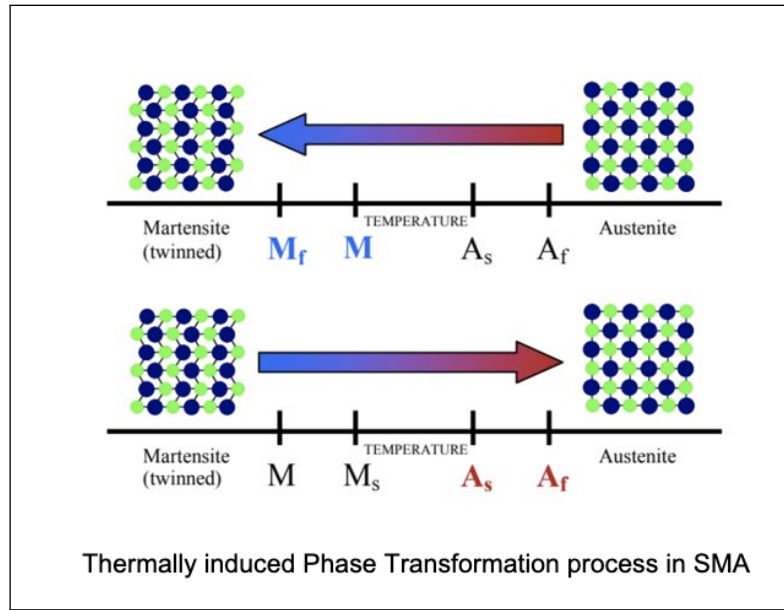


Figure 3.1: Shape Memory Effect [7]

The phase transformation as seen in the figure 3.1 from martensite to austenite (heating) causes a decrease in length while the reverse process extends it back to its original length in case of using SMA wire for actuation. The SMA wire exhibits a hysteresis behavior in its relationship between temperature and strain due to non-uniformity in the phase transformations. This occurs because the phase transformation from martensite to austenite begins and ends at different temperatures than the reverse process, and the relationship is highly nonlinear. below figures demonstrate this behavior [7]

3.2 Modeling the Shape Memory Materials' behaviour

Many of the phenomena observed in SMAs is explained in terms of underlying micro-structural mechanisms and SMA behaviour, under a given thermo-mechanical loading path. It can be phenomenological described as this current level of knowledge has developed over several decades. Though largely capable of accurately modeling SMA behavior, the constitutive models available in the literature inherently depend on the accurate characterization of the material typically in the form of identifying values for material parameters. [7]

Transformation temperatures are perhaps the most definitive physical parameters of a given SMA. These temperatures govern the transformation between the austenite and martensite phases of the SMA and are unique to the material. Because the transformation between austenite and martensite is hysteretic, SMAs are characterized with four transformation temperatures starting and finishing temperatures for both the forward and reverse transformations.

Experimental method used in determining these values is differential scanning calorimetry (DSC). In this technique, an SMA specimen is held at a constant stress and a constant rate of temperature change, and the energy required to maintain this rate is recorded. Using this method, one can identify the region of phase transformation and determine the related stress-specific transformation temperatures. Another method often used to identify transformation temperatures involves observing how strain is generated as an SMA undergoes changes in temperature while under constant stress. [13]

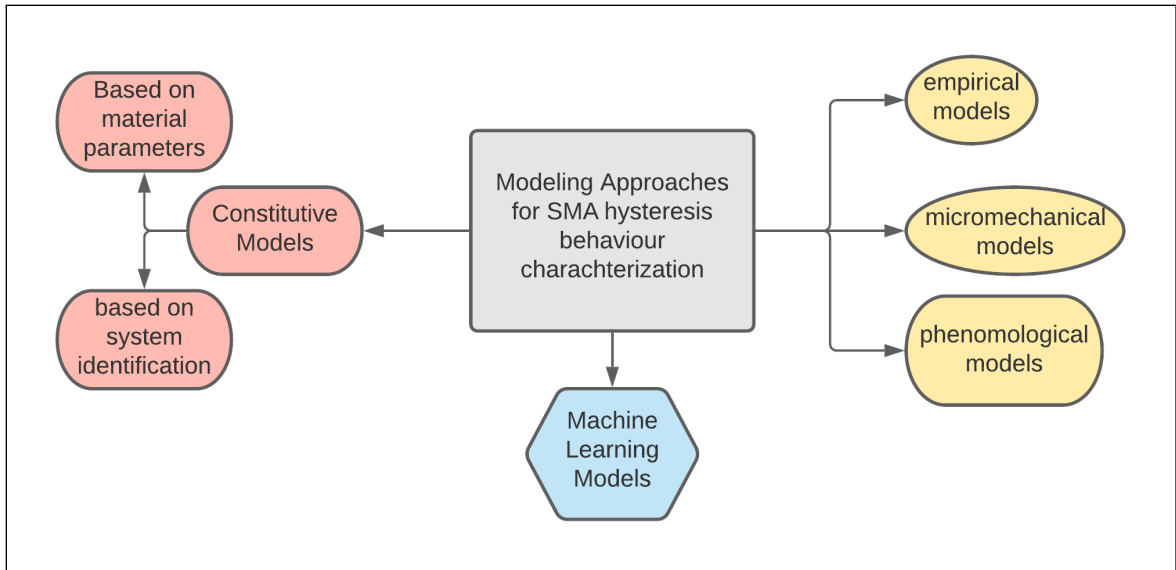


Figure 3.2: Modelling Approaches for SMA behaviour characterization

Although these methods allow one to identify SMA material properties with some degree of accuracy, these existing methods are currently time-consuming procedures requiring both extensive data collection and processing. Furthermore, these processes typically require some degree of SMA expertise. This creates a somewhat high barrier of entry for both engineers hoping to use the material and researchers desiring to further model advanced behavior of the material.

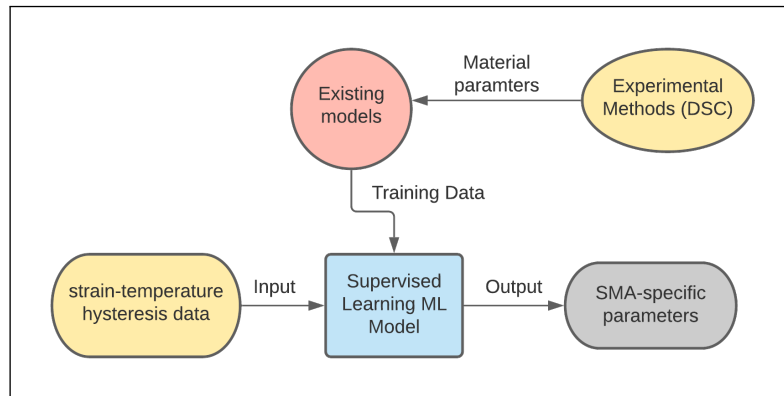
Additionally some only work for particular types of SMAs and most require complex computations. Many of them are also unable to be used in dynamic loading conditions, making them unusable in the case of morphing. A major drawback to using any of these methods is that the minor hysteresis loops that correspond to a SMA that is not fully actuated are unattainable and must be determined through mathematical models. [8]

3.3 Machine Learning based approach for SMA behaviour characterization

The problems described in the section before can be addressed with use of Machine Learning Approaches, the problem of searching for patterns in data is a fundamental one and specific developments regarding machine learning in the past few decades have included the introduction of topics such as artificial neural networks, genetic algorithms, reinforcement learning, and deep learning.

This provides a good opportunity to apply the advancements in the machine learning models for the benefit, given the nature of the approaches for the behaviour characterization, supervised learning based ML model serves as a good starting point towards the development of an accurate and effective ML model.

Supervised machine Learning model for shape memory material based actuators' behaviour characterization process summary diagram:



Considering the convenience in the final implementation of supervised machine learning models with use of python or other open source libraries, It is important to first focus on the generation of the data i.e, implementation of the existing models. Thus in the next chapter one of the constitutive model implementation approach that is, return mapping algorithm based numerical implementation of the SMA thermo-mechanical model is discussed in detail with the objective of deciding suitable return mapping algorithm for the purpose of data generation.

Chapter 4

A review of numerical implementation of SMA thermo-mechanical model

In this chapter I conducted review of numerical implementation of SMA thermo-mechanical model by qidwai et al. using return mapping algorithms with the intent to extract information relevant to its implementation for the ML model development. [14]

4.1 Introduction

The complex thermomechanical behavior of SMA components under general loading conditions were slow to be developed and integrated into finite element analysis software therefore most of the development in SMAs is result of repeated design/build/test cycles.

To address this issue of Numerical Implementation of Thermo-mechanical constitutive models, authors Siddiq et al. proposed usage of return mapping algorithms appropriate for rate-independent inelastic constitutive models. The closest point projection return mapping algorithm and the convex cutting plane return mapping algorithm are used.

Unified Thermo-mechanical constitutive model is implemented for the work by the authors. It includes exponential, polynomial and cosine model under consistent thermo-mechanical framework, with infinite strain constraint, Thus only applicable for structures going for small strains and finite displacements

Because these return mapping algorithms are used to integrate incremental constitutive equations for rate-independent inelastic materials; In case of SMA rate-independence and loading history dependence is observed therefore this algorithm is useful and suited to be applied.

Thermo-mechanical Behaviour of SMA is defined on the basis of stress-strain relation, which includes transformation strain as internal state variable, differential evolution equation of transformation strain, and transformation function to determine the onset of phase transformation, During this implementation the tangent moduli tensors is computed which relates strain and temperature increments (input) to stress increment (output).

Here $\mathcal{L}$ is the tangent stiffness tensor and Θ is the tangent thermal moduli tensor.

$$d\sigma = \mathcal{L} : d\epsilon + \Theta dT \quad (4.1)$$

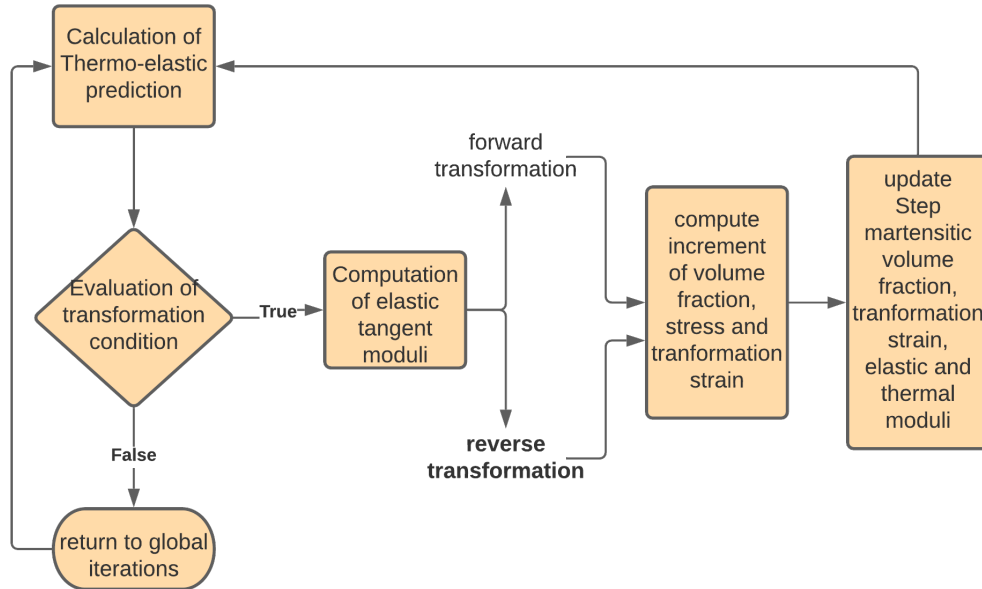
4.2 Return mapping algorithms

Return Mapping algorithm includes a purely thermo-elastic trial phase plastic corrector phase (which is transformation phase in case of SMA), the corrector phase enforces the prescribed transformation rule. These algorithms differ based on kind of discretization employed for the numerical integration and the numerical procedure used for solving the non-linear algebraic evolution equations.

4.2.1 Closest point projection return mapping algorithm

The key feature of the closest point projection algorithm is it integrates the transformation equation for the correction using the backward euler integration rule ($\lambda = 1$) resulting in non-linear algebraic equation which are solved using newton's method.

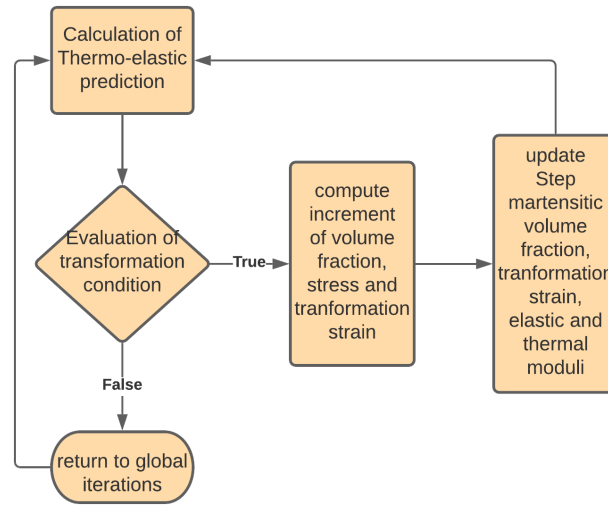
The Key steps of the algorithm are:



4.2.2 Convex cutting plane return mapping algorithm

Integration of transformation correction is done in explicit manner with linearizing the consistency condition. The algorithm converges to final value in a quadratic rate but convergence is not guaranteed. No transformation residual is required to be computed.

The Key steps of the algorithm are:



4.2.3 Comparison of both algorithms

Closest point projection

1. It is based on backward Euler integration rule, involves newton's iteration method for solving the non-linear algebraic equations generated during the transformation correction step
2. The quadratic convergence rate is guaranteed, backward Euler projections makes local integration procedure unconditionally stable and first order accurate
3. Derivation of consistent closed form tangent moduli tensor preserving global quadratic convergence rate
4. Solution of system of linear algebraic relations is required resulting into large number of tensorial operations, computation of gradient of transformation tensor and inversion of 6x6 tangent tensor during forward transformation adding to lot of computation effort and complexity

Convex cutting plane algorithm

1. Based on explicit integration of transformation evolution equation from the thermoelastic prediction state to final state
2. Forward Euler integration rule is applied and transformation evolution equations are obtained
3. In forward transformation the algorithm is unconditionally stable while for the reverse transformation newton method is locally convergent hence may result in instability
4. Derivation of consistently closed tangent moduli is not possible in this case continuum tangent moduli tensors to be used for finding incremental stress which may require a quasi-newton method for faster convergence in the global solution scheme
5. Fundamental advantages are simplicity and lower computational effort

This chapter presented methods for implementing thermomechanical model of shape memory alloys using return mapping algorithms. Two return mapping algorithms were discussed in detail, Depending on the objective and requirements of the implementation particular return mapping algorithm can be applied.

Chapter 5

Case Study: Artificial Neural Network Model for SMA characterization

In this chapter I have conducted case study of development of Artificial Neural Network based Machine learning model for Shape Memory characterization by James henrickson [8].

5.1 Artificial Neural Network Algorithm

ANN - artificial neural network is a machine learning implementation inspired by biological neural networks, which consists of a complex three-dimensional network of interconnected neurons.

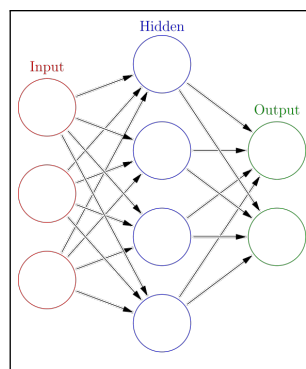


Figure 5.1: Typical ANN Structure

The pattern recognizing functionality is exhibited by modifying the strengths of the connections between the individual simulated neurons or nodes through the training process, the goal of the training process is to yield a non-parametric, black-box model of the system based on supplied training data. Thus a trained ANN model will map the pattern between input-output.

5.2 ANN Model Implementation

SMA phenomenological model parameter characterization was carried out by [8] which means desired target values (the characteristic material parameters) are identified using input data (stress specific strain-temperature hysteresis data). Supervised Learning-based ANN is used for the purpose since the input and output can be quantified prior to training.

Rather than experimentally-derived training data, constitutive model based training data generation was selected to allow for rapid assessment of the ANN approach

Implementation of constitutive SMA model allows for the generation of strain-temperature hysteresis plots using SMA parameters as inputs, the desired value of the trained ANN is to reverse this process. the target values necessary for training are obtained by simply storing the desired parameters used to obtain hysteresis plot

5.3 Results

ANN-based Approach developed by [8] is capable of identifying material parameters the transformation temperature and stress influence coefficients, with an accuracy of 87% for transformation temperature and 73% with stress influence coefficients, with the use of 428 specimens for training.

One of the key findings includes the implementation of the Taguchi method which allowed the development of a high-performance ANN model with better accuracy using 428 specimens compared to ANN trained with 4296 specimens.

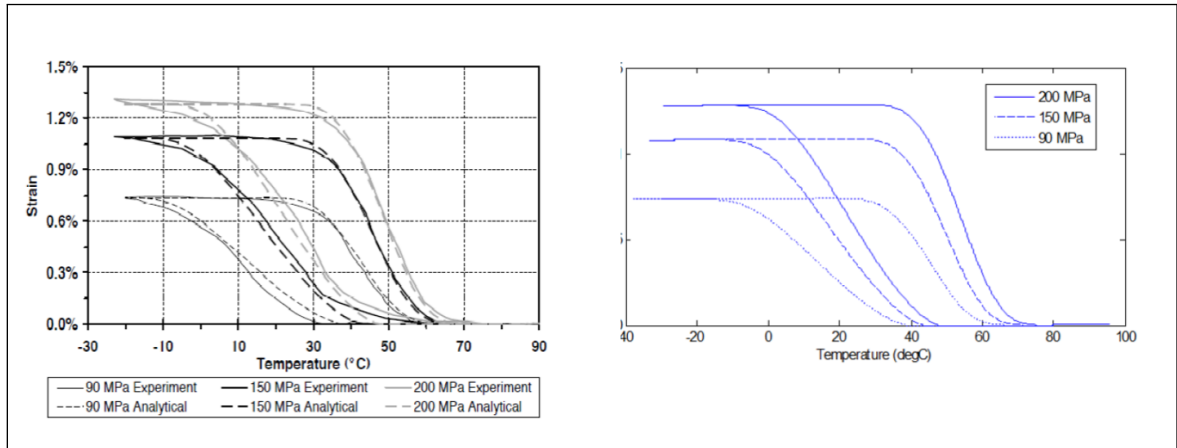


Figure 5.2: Model validations - Analytical and experimental (top) versus ANN implementation example (bottom) [8]

Once trained, this ANN model by henrickson is observed to accurately identify the SMA parameter values for a given specimen using only strain-temperature hysteresis data. This turns out to be more rapid approach to SMA material parameter identification than existing characterisation methods.

Chapter 6

Future Work

1. Developing supervised machine learning model based on the understanding gained through the project.
2. Performing the data generation and implementation of the ML model
3. Tuning the ML model parameters, exploring ways to improve the accuracy
4. Exploring hybrid machine learning approach for the purpose of characterization

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